

Tathagata Karmakar

K. B. Whaley  group

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<https://tathagata-karmakar.github.io/>

EXPERTISE

Quantum algorithms for classical optimization, quantum optimal control, continuous measurements and feedback, machine learning approaches in physics.

EDUCATION

2024 Ph.D., Physics and Astronomy, University of Rochester.
2020 M.A., Physics and Astronomy, University of Rochester.
2018 BS, Physics CPI: 9.9/10, IIT Kanpur.

PROFESSIONAL APPOINTMENTS

2024–Ongoing Postdoctoral scholar, University of California, Berkeley.
2021–2024 Affiliated student researcher, Chapman University.
Jul.–Sep. 2023 Research Intern, PHI Lab, NTT Research, Inc., CA.
May–Jul. 2017 Summer research assistant, CCA, Simons Foundation.

RECENT PUBLICATIONS

Y. Zhang, A. Sarlette, P. Lewalle, **T. Karmakar**, Thilo Scharnhorst, and K. B. Whaley, “Optimal schedule of multi-channel quantum Zeno dragging with application to solving the k-SAT problem”, *Quantum* **10**, 2167 (2026).

T. Karmakar, P. Lewalle, Y. Zhang, and K. B. Whaley, “Noise-Canceling Quantum Feedback: non-Hermitian Dynamics with Applications to State Preparation and Magic State Distillation”, *Phys. Rev. A* **113**, 042403 (2026).

T. Karmakar and A. N. Jordan, “CDJ-Pontryagin Optimal Control for General Continuously Monitored Quantum Systems”, *Quantum* **10** 2043, (2026).

Sethuraj K. R., **T. Karmakar**, S. A. Wadood, A. N. Jordan and A. N. Vamivakas, and “Experimental realization of supergrowing fields”, *Phys. Rev. Research* **6**, L032043 (2024).

T. Karmakar, P. Lewalle, and A. N. Jordan, “Stochastic path-integral analysis of the continuously monitored quantum harmonic oscillator”, *PRX Quantum* **3**, 010327 (2022).

RESEARCH EXPERIENCE

2024–Ongoing **Measurement-driven quantum algorithm:**

- Developed measurement-driven quantum k -SAT solvers and designed scalable numerical pipelines capable of simulating 20-qubit dynamics.
- Designed real-time optimal control protocols for measurement-driven computation, improving query complexity.

2024–2025 **Noise-canceling feedback:**

- Designed feedback protocols to generate deterministic dynamics in continuously monitored systems.
- Showed that noise-canceling feedback leads to a 300-400% boost in successful magic state distillation probabilities with the $[[5, 1, 3]]$ code.

- 2023–2025* **Quantum optimal control:**
- Generalized Pontryagin maximum principle to find the optimal control for general continuously monitored systems.
 - Designed optimal control protocols that lead to a 40-190% increase in the state preparation success rates.
- 2023–2024* **ML-based Model reduction in nonlinear optics, NTT Research, Inc:**
- Built a physics-informed neural operator-based learning architecture that solves unitary dynamics for quantum harmonic oscillators.
- 2020–2021* **Stochastic path integral:**
- Formulated a stochastic action principle-based description of the optimal evolution of continuously monitored harmonic oscillators.

RECENT TALKS

- June 2026* *Optimal control in monitored quantum systems with applications to quantum algorithms for solving classical optimization problems, CIQC, Pacific Grove, CA, USA.*
- May. 2026* *Optimal control in monitored quantum systems with applications to quantum algorithms for solving classical optimization problems, TIFR Hyderabad, Hyderabad, India.*
- May. 2026* *Convergence and schedule optimization of multi-channel quantum Zeno dragging with application to k-SAT problems, UCSB, Santa Barbara, CA.*
- Mar. 2026* *Convergence and schedule optimization of multi-channel quantum Zeno dragging with application to k-SAT problems: part II, APS GPS, Denver, CO.*
- Sep. 2025* *Invited: CDJ-Pontryagin Optimal Control for General Continuously Monitored Quantum Systems, Chapman University, Orange, CA.*
- Jun. 2025* *Noise-Canceling Feedback for Continuously Monitored Systems, CQS-12, Rochester, NY.*

PROGRAMMING EXPERIENCE

Python (PyTorch, JAX, Numba), Mathematica , QuTiP.

AWARDS AND FELLOWSHIPS

- 2020* Okubo Prize, Department of Physics and Astronomy, UR.
- 2018-2020* Robert L. and Mary L. Sproull fellow, UR.
- 2017* S. N. Bose Scholar (WSF, DST Govt. of India, IUSSTF).
- 2016 & 2015* Academic Excellence Award (dean's office, IIT Kanpur).
- 2014-2018* KVPY fellow, DST, Govt. of India.

TEACHING EXPERIENCE

- Jun.–Jul. 2026* QCaMP Educators and students, QSA, LBNL.
- Jan.–Apr. 2019* Teaching assistant, 20th Century Physics.
- Aug.–Nov. 2018* Teaching assistant, Gravitation and General Relativity.